part 1: from https://www.kaggle.com/datasets , I choose Predicting whether a customer will churn database to work with.
https://www.kaggle.com/code/emineyetm/telco-customer-churn

part 2:

• Is this a regression, classification, or clustering problem?

-it is classification

• Is there a target variable?

-yes

• What is the model expected to learn or predict?

-is the customer will churn?yes or no.

part 3:

```
# Step 1: Import Libraries
import pandas as pd
import numpy as np

# Step 2: Load the Dataset
try:
    df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')
    print("Dataset loaded successfully!")
except FileNotFoundError:
    # Fallback to load from an online repository if not found locally
    df = pd.read_csv('https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv')
    print("Dataset loaded from GitHub repository!")

# Step 3: Display the Shape of the Dataset
print("Dataset shape:", df.shape)

# Step 4: Preview the First Few Rows
print("\nFirst 5 rows:")
print(df.head())

# Step 5: Check Column Names and Data Types
print("\nColumns:", df.columns.tolist())
print("\nData Types:\n", df.dtypes)

# Step 6: Check the Target Variable Distribution
print("\nChurn Distribution:\n", df['Churn'].value_counts())

# Step 7: Fix TotalCharges Data Type and Check for Missing Values
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
print("\nMissing Values after conversion:\n", df.isnull().sum())
```

```
Dataset loaded from GitHub repository!
Dataset shape: (7043, 21)

First 5 rows:
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

                PaymentMethod MonthlyCharges  TotalCharges Churn
```

```
0           Electronic check          29.85         29.85    No
1               Mailed check          56.95        1889.5    No
2               Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4           Electronic check          70.70        151.65   Yes

[5 rows x 21 columns]

Columns: ['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'Interne

Data Types:
 customerID          object
gender               object
SeniorCitizen         int64
Partner              object
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges         object
```

Part 4:

Start coding or generate with AI.

Customer Churn Prediction